

Serverless & Modern Web Hosting

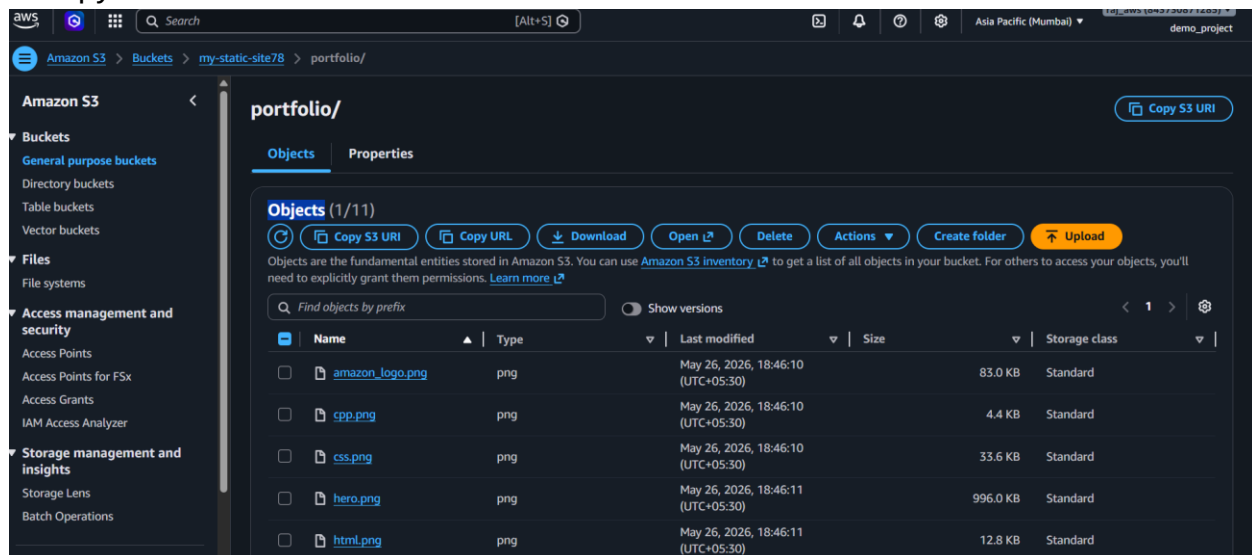
Project 1: Serverless Static Website – Use S3, CloudFront, and Lambda@Edge for global performance.

A serverless static website on AWS eliminates the need for traditional web servers by using Amazon S3 for hosting static content, CloudFront for global content delivery, and Lambda@Edge for dynamic content processing. This setup provides high availability, low latency, and cost efficiency.

Steps to Deploy a Serverless Static Website on AWS

Step 1: Create an S3 Bucket for Website Hosting

1. Open the AWS Management Console → Navigate to S3.
2. Click Create bucket, enter a unique name, and choose the region.
3. Uncheck "Block all public access" (if you want a public website).
4. Upload your static website files (HTML, CSS, JS, images).
5. Enable Static website hosting under Properties:
 - Choose "Enable".
 - Set index.html as the default document.
 - Copy the Bucket Website URL.



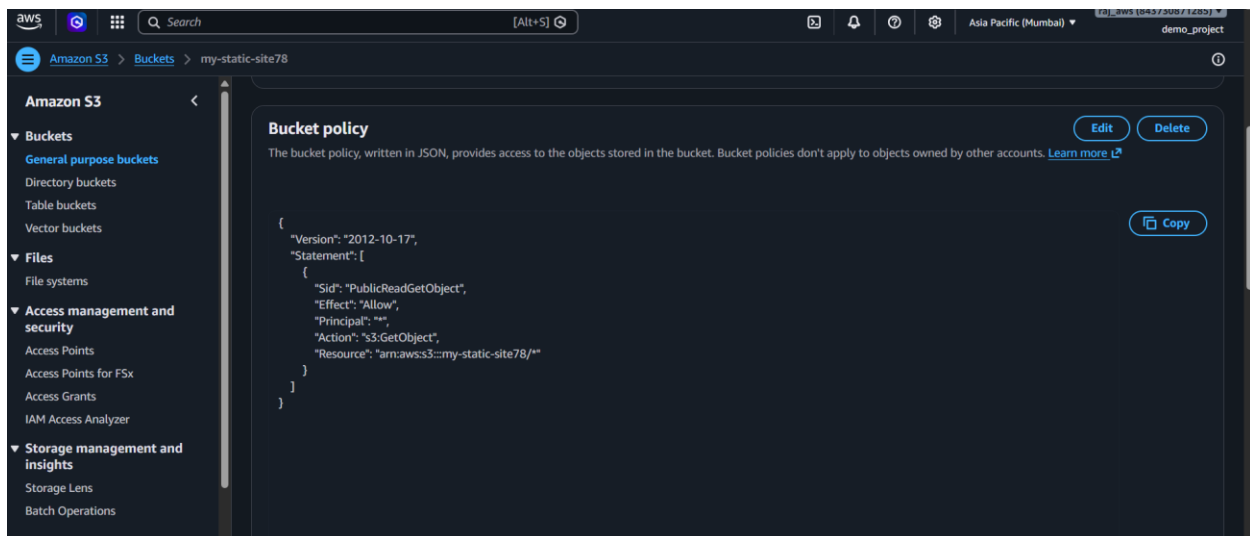
Step 2: Configure Bucket Policy for Public Access

1. Go to Permissions → Click Bucket Policy

Add the following policy to make the website publicly accessible:

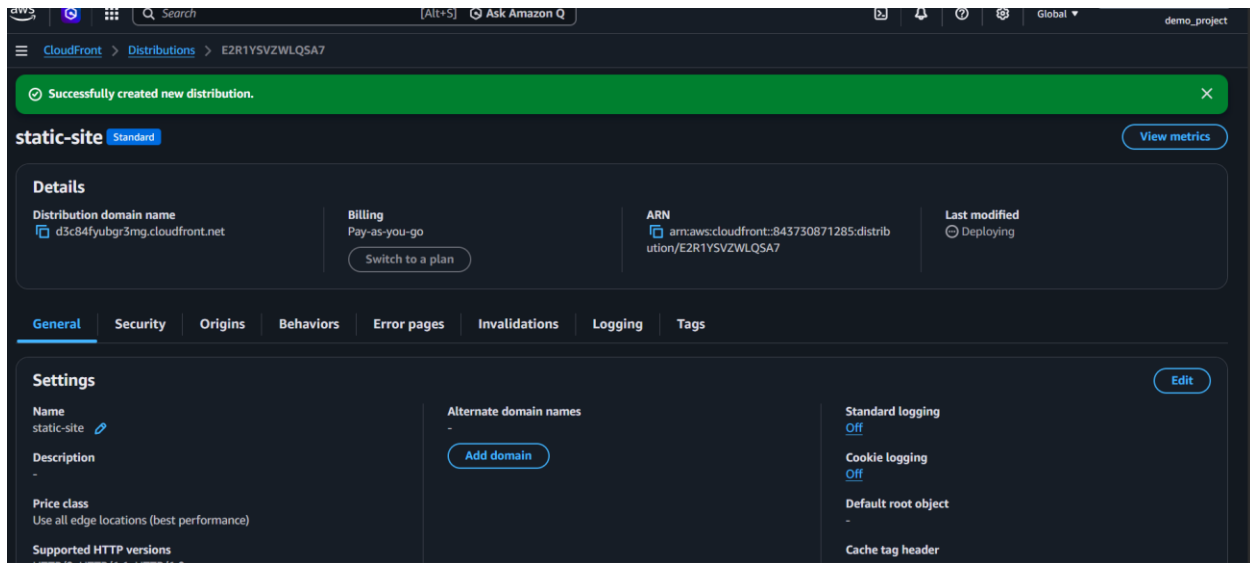
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-static-website/*"
    }
  ]
}
```

2. Save the policy.



Step 3: Create a CloudFront Distribution

1. Go to CloudFront in AWS Console.
2. Click Create Distribution.
3. Under Origin, select your S3 bucket (not the website endpoint, but the bucket itself).
4. Enable "Redirect HTTP to HTTPS".
5. Set Caching Policy to CachingOptimized (for better performance).
6. Click Create and note the CloudFront URL



Conclusion

With this setup, your static website is hosted on S3, served globally using CloudFront, and can be enhanced with Lambda@Edge for additional processing. This architecture ensures scalability, security, and cost efficiency.

